Kirshanthan Sundararajah

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Virginia Tech
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RESEARCH INTERESTS

- Compilers, Programming Languages and High-Performance Computing

EDUCATION

Purdue University, West Lafayette, IN. Aug 21, 2015 - Aug 04, 2023
Ph.D. in Electrical and Computer Engineering Aug 04, 2023

- Dissertation: *Composable, Sound Transformations of Nested Recursion and Loops.*
- Adviser: Milind Kulkarni

M.S. in Electrical and Computer Engineering Aug 05, 2022

University of Moratuwa, Katubedda, Sri Lanka. Jul 21, 2009 - Mar 01, 2014
B.Sc.(Hons) in Electronics and Telecommunication Engineering Aug 28, 2014

REFEREED PUBLICATIONS

- K. Mahathevan, P. Gunathilaka, **K. Sundararajah**. "Splyce: SIMD Vectorization of Sparse Coiteration" in *International Conference on Parallel Architectures and Compiler Techniques*, PACT 2026. [ACM DL] 🏆 **Chameleon Reproducibility Challenge Award**
- P. Gunathilaka, N. Chaudhury, **K.Sundararajah**, W. Feng. "A Preliminary Study on Simultaneous Coscheduling for Discrete GPU vs. Fused GPU" in *State of Practice in Deploying Supercomputers with NVIDIA Superchips Workshop*, SPIN-NVSC 2026. [IEEE Xplore]
- Y. Li, S. R. Sharma, C. Saumya, A. R. Butt, **K. Sundararajah**. "Eliminate Branches by Melding IR Instructions" in *European Conference on Object-Oriented Programming*, ECOOP 2026. [DROPS]
- P. Lobo, J. McIver, G. Mitenkov, J. Lee, **K. Sundararajah**, N. P. Lopes. "Towards Removing Undef Values From LLVM IR" in *ACM SIGPLAN Conference on Programming Languages, Design and Implementation*, PLDI 2026. [ACM DL] 🏆 **Distinguished Paper Award**
- C. Saumya, M. Hassan, R. Gangaraju, M. Kulkani, **K. Sundararajah**. "Taming the Hydra: Targeted Control-Flow Transformations for Dynamic Symbolic Execution" in *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA 2026. [ACM DL]
- K. Mahathevan, Y. Zhang, M. A. Gulzar, **K. Sundararajah**. "TenSure: Fuzzing Sparse Tensor Compilers" in *Fuzzing Workshop*, FUZZING 2026. [NDSS Proceedings]
- Y. Li, S. Yao, J. Mobin, T. Zhan, M. M. Rafique, D. Nikolopoulos, **K. Sundararajah**, A. R. Butt. "Memory Tiering in Python Virtual Machine" in *Virtual Machines and Intermediate Languages*, VMIL 2025. [ACM DL]
- V. Nagarajan, R. Gangaraju, **K. Sundararajah**, A. Pelenitsyn, and M. Kulkarni. "RT-BarnesHut: Accelerating Barnes-Hut Using Ray-Tracing Hardware" in *Principles and Practices of Parallel Programming*, PPoPP 2025. [ACM DL]

- A. Dias, L. Anderson, **K. Sundararajah**, A. Pelenitsyn, and M. Kulkarni. "SparseAuto: An Auto-Scheduler for Sparse Tensor Computations Using Recursive Loop Nest Restructuring" in *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA 2024. [ACM DL]
- V. Singhal, L. Sakka, **K. Sundararajah**, R.R. Newton, and M. Kulkarni. "Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals" in *Transactions on Architecture and Code Optimization*, TACO 2024. [ACM DL]
- R. C. O. Rocha, C. Saumya, **K. Sundararajah**, P. Petoumenos, M. Kulkarni, and M. F. P. O'Boyle "HyBF: A Hybrid Branch Fusion Strategy for Code Size Reduction" in *International Conference on Compiler Construction*, CC 2023. [ACM DL]
- **K. Sundararajah**, C. Saumya, and M. Kulkarni "UniRec: A Unimodular-Like Framework for Nested Recursions and Loops" in *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA 2022. [ACM DL]
- A. Dias, **K. Sundararajah**, C. Saumya, and M. Kulkarni "SparseLNR: Accelerating Sparse Tensor Computations Using Loop Nest Restructuring" in *ACM International Conference on Supercomputing*, ICS 2022. [ACM DL] 🏆 **Best Paper Award**
- C. Saumya, **K. Sundararajah**, and M. Kulkarni "DARM: Control-Flow Melding for SIMT Thread Divergence Reduction" in *IEEE Symposium on Code Generation and Optimization*, CGO 2022. [IEEE Xplore]
- Y. Bao*, **K Sundararajah***, R. Malik, Q. Ye, C. Wagner, N. Jaber, F. Wang, M. H. Ameri, D. Lu, A. Seto, B. Delaware, R. Samanta, A. Kate, C. Garman, J. Blocki, P. Letourneau, B. Meister, J. Springer, T. Rompf, and M. Kulkarni "HACCLE: Metaprogramming for Secure Multi-Party Computation" in *International Conference on Generative Programming: Concepts and Experiences*, GPCE 2021. [ACM DL]
- **K. Sundararajah** and M. Kulkarni "Composable, Sound Transformations of Nested Recursion and Loops" in *ACM SIGPLAN Conference on Programming Languages, Design and Implementation*, PLDI 2019. [ACM DL]
- L. Sakka, **K. Sundararajah**, R. R. Newton, and M. Kulkarni "Sound, Fine-Grained Traversal Fusion for Heterogeneous Trees" in *ACM SIGPLAN Conference on Programming Languages, Design and Implementation*, PLDI 2019. [ACM DL]
- L. Sakka, **K. Sundararajah** and M. Kulkarni "TreeFuser: A Framework for Analyzing and Fusing General Recursive Tree Traversals" in *Object-Oriented Programming, Systems, Languages, and Applications*, OOPSLA 2017. [ACM DL]
- N. Hegde, J. Liu, **K. Sundararajah**, and M. Kulkarni "Treelogy: A Benchmark Suite for Tree Traversals" in *IEEE International Symposium on Performance Analysis of Systems and Software*, ISPASS 2017. [IEEE Xplore]
- **K. Sundararajah**, L. Sakka, and M. Kulkarni "Locality Transformations for Nested Recursive Iteration Spaces" in *ACM International Conference on Architectural Support for Programming Languages and Operating Systems*, ASPLOS 2017. [ACM DL]
- **K. Sundararajah** and S. Jayasena, "Model-based Input-adaptive Vectorization" in *Moratuwa Engineering Research Conference*, MERCon 2016. [IEEE Xplore]
- **K. Sundararajah**, L. Logeswaran, P. N. D. Panagoda, L. P. Wijesinghe, D. V. S. X. De Silva, and A. A. Pasqual, "Layered Depth Image Based HEVC Multi-view Codec" in *Advances in Visual Computing: Proceedings of the International Symposium on Visual Computing*, ISVC 2014. [Springer]

PROFESSIONAL EXPERIENCE

- **Assistant Professor of Computer Science** @*Virginia Tech* Aug 10, 2023 - Present
- **Graduate Research Assistant** @*Purdue University* Aug 21, 2015 - Jul 28, 2023
  PLCL Group
  - **(Poly/Uni)Rec**: Framework for composing irregular program transformations.[PLDI'19, OOPSLA'22]
  - **SparseLNR**: Framework for accelerating sparse tensor computation.[ICS'22]
  - **DARM/HyBF**: Framework for melding similar control-flow graphs.[CGO'22, CC'23]
  - **HACCLE**: Ecosystem for *Secure Multi-Party Computations*.[GPCE'21]
  - **TreeFuser/Grafter**: Framework for fusing general recursive traversals. [OOPSLA'17, PLDI'19]
  - **Treelogy**: Benchmark suite for tree traversals.[ISPASS'17]
  - **Recursion Twisting**: Optimizing nested recursive traversals.[ASPLOS'17]
- **Software Engineering Intern** @*Nvidia* Sep 21, 2020 - Dec 18, 2020
  GPU Compiler Group
  - **Diesel Compiler**: Warp specialization and pipelining for GPU kernels.
- **Software Engineering Intern** @*Reservoir Labs (Now Qualcomm AI Research)* Jun 01, 2020 - Aug 28, 2020
  R-Stream Compiler Group
  - **ParSEC Backend**: A task-based runtime backend for *R-Stream* polyhedral compiler.
- **Research Intern** @*Microsoft Research* Jun 11, 2018 - Sep 14, 2018
  RiSE Group
  - **Parallelizing Word2Vec**: Parallelizing and scaling *Word2Vec* training to execute on many cores.
- **Research Assistant** @*University of Moratuwa* Mar 02, 2014 - Jul 25, 2015
  Department of CSE
  - **Vectorization**: Model-based Input-adaptive Vectorization
- **Associate Electronic Engineer** @*Zone24x7 Inc.* May 14, 2012 - Oct 12, 2012
  Signs24x7 Group
  - **Image Compression Algorithm**: Implementation of memory efficient image compression algorithm, supposed to perform decompression on an *STM32 microcontroller* based system.
  - **Clock Synchronization Algorithm**: Implementation of real-time clock synchronization algorithm, deployed on an *ARM microprocessor* runs *embedded Linux*.
  - **Hardware Abstraction Layer**: Implementation of *Hardware Abstraction Layer (HAL)* for radio communication protocol stack of *Electronic Paper Display(EPD)*, driven by an *STM32 microcontroller*.

AWARDS

- *Chameleon Reproducibility Challenge Award* at PACT 2026.
- *Distinguished Paper Award* at PLDI 2026.
- *Best Paper Award* at ICS 2022.

- *Bilsland Dissertation Fellowship 2021-2022*, Purdue University.
- *Silver Medal* in ACM Student Research Competition at *SPLASH 2018*.
- *Electrical and Computer Engineering Fellowship 2015-2016*, Purdue University.
- *V. K. Samaranayake Research Assistantship 2014-2015*, University of Moratuwa.
- *Mahapola Merit Scholarship 2009-2014*, University of Moratuwa.

TEACHING

- CS 3114 Data Structures and Algorithms @*Virginia Tech* Fall 2026
- CS 4114 Formal Languages @*Virginia Tech* Sping 2026
- CS 4304/5304 Compiler Design @*Virginia Tech* Fall 2025
- CS 5134 Programming Languages @*Virginia Tech* Fall 2024
- CS 6304 Advanced Topics in Language and Translator Design @*Virginia Tech* Spring 2024
- CS 4304/5304 Compiler Design @*Virginia Tech* Fall 2023
- ECE 368 Data Structures @*Purdue University* Summer 2021 and Summer 2022
- ECE 295 Introduction to Data Science [TA] @*Purdue University* Summer 2019, Fall 2020, and Summer 2023
- CS 1032 Programming Fundamentals [TA] @*University of Moratuwa* Mar 2014 - Jun 2015

SERVICE

- **Organizing**
  - Program Chair for SRC at CGO 2026.
  - Program Committee Chair VMIL 2026.
  - Registration Chair for PPoPP 2023.
  - Member of Project for Inclusion in ECE (PIECE) Committee 2022.
  - Student volunteer for PLDI 2016, SPLASH 2018, PLDI 2019, and SPLASH 2021.
  - Co-organizer of PurPL weekly seminar (Fall 2017 - Spring 2021) and volunteer for PurPL Fest 2019.
- **Reviewing**
  - Member of Program Committee (PC) IPDPS 2027 and ICS 2027.
  - Member of Program Committee (PC) PPoPP 2026, OOPSLA 2026, and ICPP 2026.
  - Member of Program Committee (PC) for CGO 2025, PPoPP 2025, PLDI 2025, PLDI 2025 SRC.
  - Member of Program Committee (PC) for PLDI 2024 SRC and CC 2024.
  - Member of External Review Committee (ERC) for OOPSLA 2022.
  - Member of Artifact Evaluation Committee (AEC) for OOPSLA 2022.
  - Member of Artifact Evaluation Committee (AEC) for PLDI 2020.
  - Member of Program Committee (PC) for Doctoral Symposium at ECOOP 2019.
  - External collaborative reviewer for POPL 2019.

STUDENTS

- Muhammad Hassan (PhD @VT) Jan 2025 - Present
  Applications of Bidirectional Control-flow Transformations.
- Kabilan Mahathevan (PhD@VT) Aug 2025 - Present
  Correctness and Performance of Sparse Tensor Compilers.
- John McIver (PhD@VT) Aug 2025 - Present
  Generalizing Fusion Transformations in Compilers.
- Poorna Gunathilaka (PhD@VT) Aug 2025 - Present
  CPU-GPU Coscheduling for HPC systems.
- Yining Zhang (PhD@VT) Aug 2025 - Present
  Automatic Specialization of Sparse Tensor Compilers.
- Srinivasan Ramachandra Sharma (MS @VT → Qualcomm) Jan 2024 - Jul 2025
  Compiler Transformation to Mitigate Branch Misprediction by Branch Elimination.
- Yuze Li (PhD @VT: Advised by Ali Butt → Huawei) Sep 2023 - May 2026
  Cross-stack Improvement on Memory Efficiency.
- Wei-Jhe Lin (UG@VT) Jan 2026 - May 2026
  Register Tiling for MatMul Units in GPUs.

ACHIEVEMENTS

- **Grants**
  - ACM Travel Grant to Attend *PLDI 2019*, *SPLASH 2018*, *ASPLOS 2017*, *PLDI 2016*, and *CGO 2015*.
- **Competitions**
  - Placed 25th, 34th, 29th, and 45th in *IEEExtreme 7.0, 6.0, 5.0 and 4.0*, respectively.
  - Placed 4th in *Sri Lanka Robot Competition (SLRC) 2012*.
  - Champions of *Inter-University Statistics Quiz Competition 2010*, University of Colombo, Sri Lanka.
  - Participated at *International Mathematics Olympiad Competition(IMO) 2009*, Bremen, Germany